

07 - Clustering Evaluation

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Introduction

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 - Rand
 - Jaccard
 - Fowlkes and Mallows
 - Gamma
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 - Jaccard
 - Fowlkes and Mallows
 - Gamma
- Internal Validation metrics
 - Choesion
 - Separation
 - Silhouette Coefficient



External Measures

External/Supervised validation

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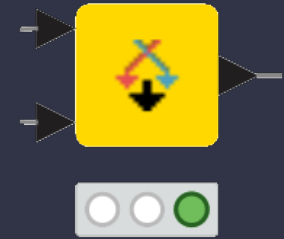
For each pair of records in the dataset we can identify 4 cases

- **Same** cluster - **Same** class
- **Same** cluster - **Different** class
- **Different** cluster - **Same** class
- **Different** cluster - **Different** class

Cross Joiner

The *Cross Joiner* node perform a **cross join** where **each row** of the **top** table is **joined** with **each row** of the **bottom** table.

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Table A

A	B
1	0
12	3

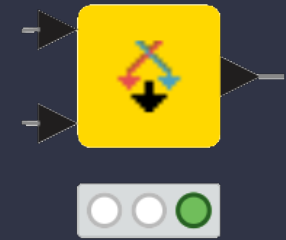
Table B

A	C
2	0
12	4

Cross Join

A	B	A1	C
1	0	2	0
1	0	12	4
12	3	2	0
12	3	12	4

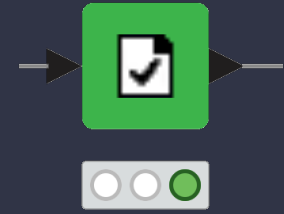
Cross Joiner



Rule Engine

The *Rule Engine* node takes a **list of user-defined rules** and tries to **match** them to each row in the **input table**. The **first matching rule** in order of definition **determines the outcome**.

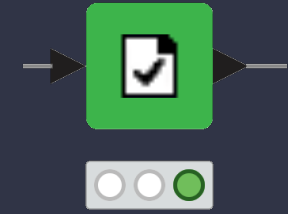
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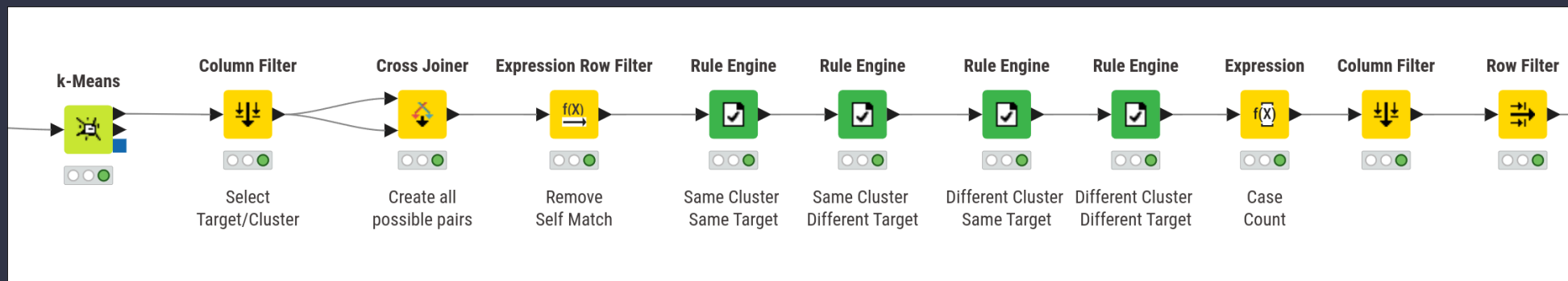


The screenshot shows the 'Rule Editor' window with the following components:

- Column List:** A list of available columns including ROWID, ROWINDEX, ROWCOUNT, target, Cluster, target (#1), Cluster (#1), FirstRowIDs, and SecondRowIDs. The 'target' and 'Cluster' columns are selected.
- Flow Variable List:** A list containing 'knime.workspace'.
- Function:** A list of logical functions including '<', '<=', '=', '>', '>=', 'AND', 'IN', 'LIKE', 'MATCHES', 'OR', and 'XOR'.
- Expression:** A text area containing a set of rules:

```
1 // enter ordered set of rules, e.g.:
2 // $double column name$ > 5.0 => "large"
3 // $string column name$ LIKE "*blue*" => "small and blue"
4 // TRUE => "default outcome"
5 $target$ = $target (#1)$ AND $Cluster$ = $Cluster (#1)$ => 1
6 TRUE => 0
```
- Append Column:** A radio button selected, with a text field containing 'Same Cluster Same Target'.
- Replace Column:** A radio button unselected, with a dropdown menu showing 'SecondRowIDs'.

Compute External Validation Statistics

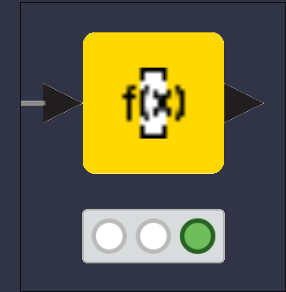


External Validation Metrics

Let:

- a : Count(**Same** cluster - **Same** class)
- b : Count(**Same** cluster - **Different** class)
- c : Count(**Different** cluster - **Same** class)
- d : Count(**Different** cluster - **Different** class)
- $M = \frac{m \times (m-1)}{2} = a + b + c + d$
 - m is the number of records

Expression

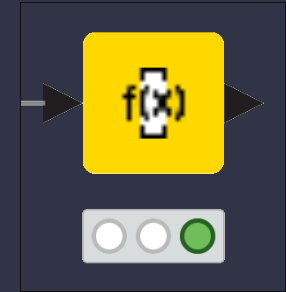


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Rand

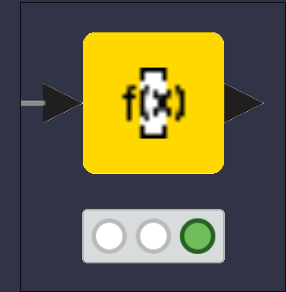
$$R = \frac{a + d}{M} \quad R \in [0, 1]$$

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Jaccard

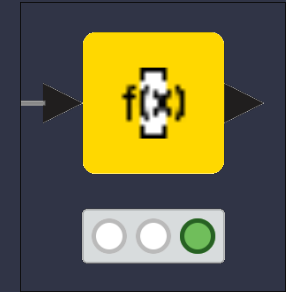
$$J = \frac{a}{a + b + c} \quad J \in [0, 1]$$

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Expression



Fowlkes And Mallows

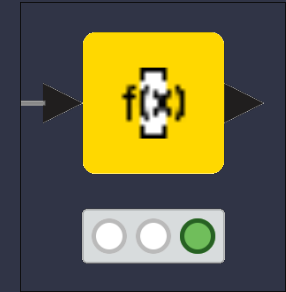
$$FM = \sqrt{\frac{a}{a+b} \times \frac{a}{a+c}} \quad FM \in [0, 1]$$

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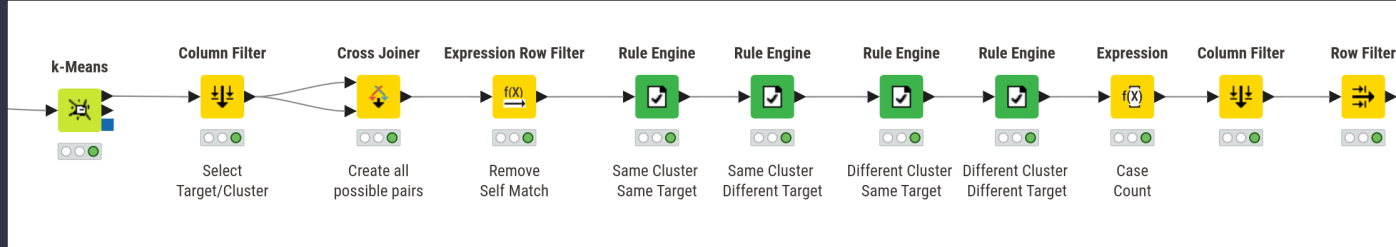


Γ statistics

$$\Gamma = \frac{M \times a - (a + b) \times (a + c)}{\sqrt{(a + b) \times (a + c) \times (c + d) \times (b + d)}} \quad \Gamma \in [-1, 1]$$

Hands on (30 minutes)

- Load the **iris** dataset
 - Use the *python* node
 - Otherwise, load the *iris.csv*
- Plot the dataset using the *Color Manager* node in combination with the *Scatter Matrix* node.
- Use any cluster algorithm
- Extract the External Validation Statistics:



- Compute the following external metrics:
 - Rand
 - Jaccard
 - Fowlkes And Mallows
 - Γ statistics

Internal Measures

Cohesion

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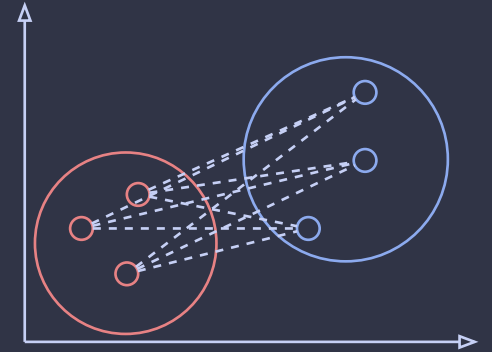
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Higher values of **cohesion** are **better**.



Separation

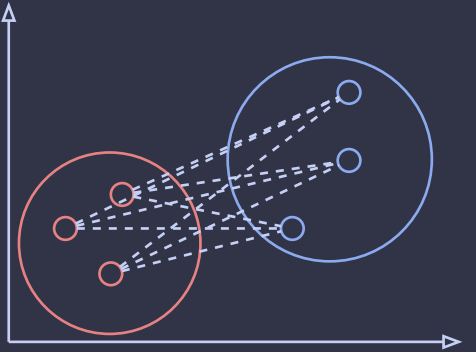
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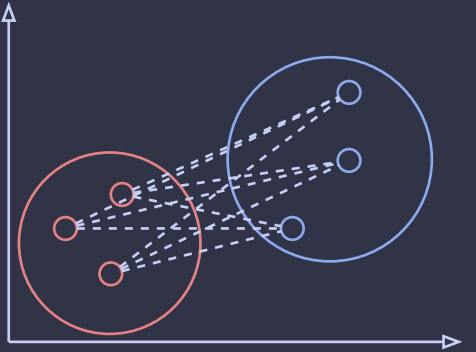
$$\text{separation}(C_i, C_j) = \sum_{x \in C_i, y \in C_j} \text{proximity}(x, y) \approx \frac{1}{\sum_{x \in C_i, y \in C_j} \text{distance}(x, y)}$$



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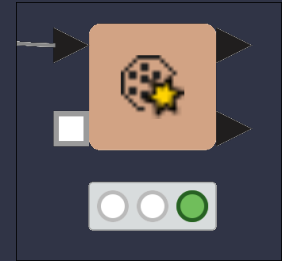


Lower values of **separation** are **better**.

Silhouette Coefficient

The **Silhouette Coefficient** is an internal cluster evaluation measure.

**Silhouette
Coefficient**



Silhouette Coefficient

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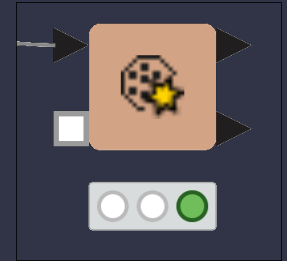
$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

where:

$$P_i = \frac{1}{|P_i|} \sum_j \text{distance}(i, j) \quad i, j \in P_i$$

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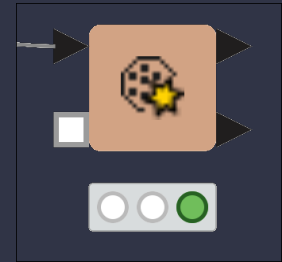
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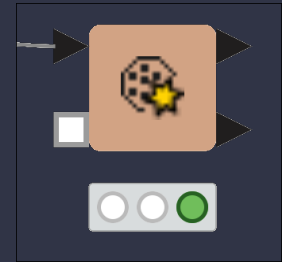
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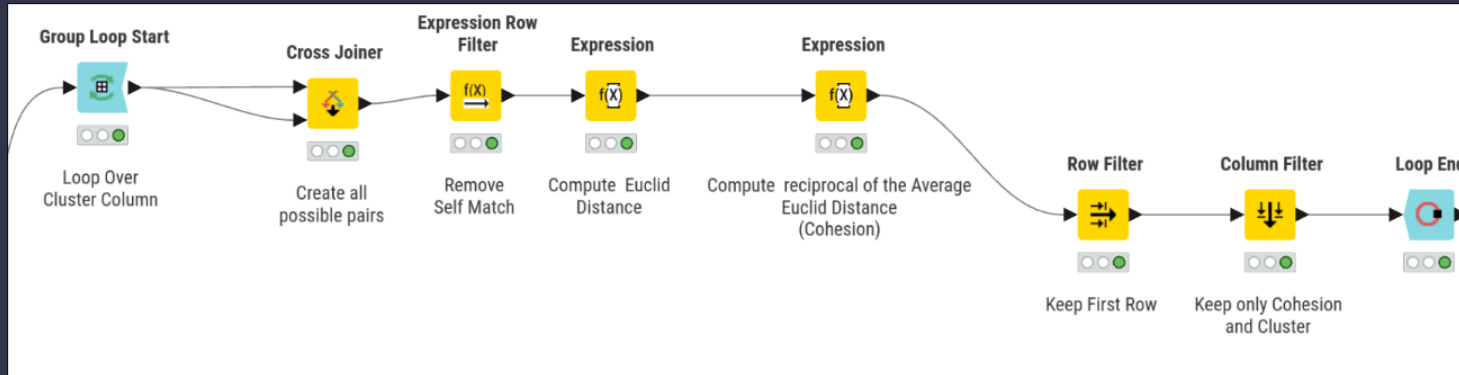
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Silhouette Coefficient



Hands on (40 minutes)

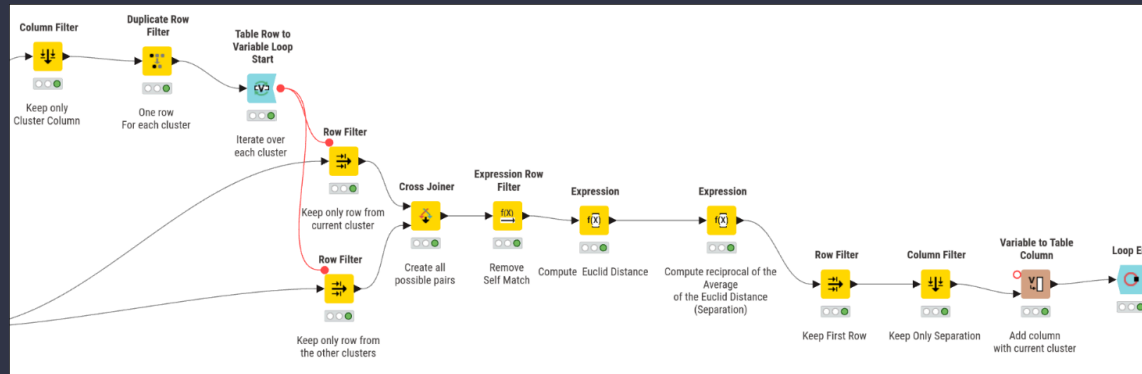
- **Extend** the previous workflow.
- **Implement Cohesion:**



- **Implement Separation**
- **Compute** the Silhouette coefficient
- **Train a clustering algorithm** with 2,3,4,5 clusters
- Use both **external** and **internal** measure to identify the optimal number of clusters.
- **Plot** the metrics

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